Fiscal Year:	FY 2004	**Task Last Updated:**	FY 04/01/2006
PI Name:	Potember, Richard S Ph.D.		
Project Title:	Real-Time Analysis of Biomarkers and Countermeasures Usinmg a Miniature Time of Flight Mass Spectrometer		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Technology Development Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	redwing@aplcomm.jhuapl.edu	**Fax:**	FY 240-228-6904
PI Organization Type:	UNIVERSITY	**Phone:**	240-228-6251
Organization Name:	The Johns Hopkins University		
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PI Web Page:	http://www.apl.jhu.edu/		
City:	Laurel	**State:**	MD
Zip Code:	20723	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NSBRI
Start Date:	11/01/2000	**End Date:**	12/31/2005
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Start/end dates corrected to 11/1/2000-12/31/2005 per K. Major/NSBRI (previously 11/1/2001-12/31/2005)--2/2009		
Key Personnel Changes/Previous PI:	Potember, Richard		
COI Name (Institution):			
Grant/Contract No.:	NCC 9-58-TD.002.05		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The aim of this task is to demonstrate that the miniature TOF system is capable of detecting and quantifying different circulating stress biomarkers that appear during space flight. These biomarkers are a measure of muscle damage (stress). Detection and quantification of these markers using the miniature TOF technology will allow real-time monitoring of muscle damage on-orbit.		
Rationale for HRP Directed Research:			

Research Impact/Earth Benefits:	The objective of this ground-based advanced technology development project is to lay the scientific and engineering foundations to design, build and launch a flight qualified “Miniature Time-of-Flight Mass spectrometer” (TOFMS) for use on space platforms such as the Space Shuttle, the International Space Station (ISS) and a human mission to Mars. The development of a “Miniature Time-of-Flight (TOF) Mass Spectrometer” will provide the NSBRI/NASA with a complete medical diagnostic system to monitor human physiological functions (Type I, II and III risks as identified in NASA’s Critical Path Roadmap) routinely and non-invasively. This compact medical diagnostic system will provide autonomous and semi-autonomous patient monitoring systems with low false positive alarm rates. The instrument will be compatible with expert medical systems and telemedicine procedures. This instrument may also be used to detect and identify pathogen that may be present on the space station.
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)